

Experiment ‘p_2_3_power_functions’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_3_power_functions

Natural language statement: null

Method signature: p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)

Ensures

- `sq == x * x`
- `cube == x * x * x`
- `fourth == x * x * x * x`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)`

The method should respect the following contract:

`ensures sq == x * x, ensures cube == x * x * x, ensures fourth == x * x * x * x`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2gh2w7Vfkgqs1Mu8ENi0drmJSgL

Model: gpt-5.5-2026-04-23

Created at: 1780580775

Finish reason: stop

Usage: 300 tokens in, and 38 tokens out

```
{
  sq := x * x;
  cube := sq * x;
  fourth := sq * sq;
}
```

Final Program

```
method p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)
  ensures sq == x * x
  ensures cube == x * x * x
  ensures fourth == x * x * x * x
{
  sq := x * x;
  cube := sq * x;
  fourth := sq * sq;
}
```

Total Token Usage

Input tokens: 300

Output tokens: 38

Reasoning tokens: 0

Sum of 'total tokens': 338

Experiment Timings

Overall Experiment started at 1780580773209, ended at 1780580799043, lasting 25834ms (25.83 seconds)

Iteration #1 started at 1780580773214, ended at 1780580799043, lasting 25829ms (25.83 seconds)